CIFAR-10 Image Classification using CNNs

Part (B): VAE-Based Data Augmentation

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1 Overview

This section describes the use of a Conditional Variational Autoencoder (CVAE) to generate synthetic training images for each of the 10 CIFAR-10 classes, and evaluates whether augmenting the CNN classifier from Part (a) with these generated samples improves classification performance.

2 Methodology

2.1 CVAE Training

The CVAE was trained minimising a combined reconstruction and KL divergence loss. The reconstruction term measures mean squared pixel-wise error between input and output:

$$\mathcal{L}_{\text{recon}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{D} (x_{ij} - \hat{x}_{ij})^2$$

and the KL term regularises the latent space toward a standard normal distribution:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2N} \sum_{i=1}^{N} \sum_{k=1}^{L} \left(1 + \log \sigma_{ik}^2 - \mu_{ik}^2 - \sigma_{ik}^2\right)$$

KL annealing was applied to prioritise reconstruction before latent regularisation. Total loss decreased from 102.44 at epoch 1 to 72.63 at epoch 60, indicating stable convergence and a well-balanced trade-off between reconstruction fidelity and a well-structured latent space suitable for sampling novel images.

Table 1: CVAE Hyperparameters

Parameter	Value
Latent dim	256
Epochs	60
Batch size	128
Epoch 1 loss	102.44
Final loss	72.63

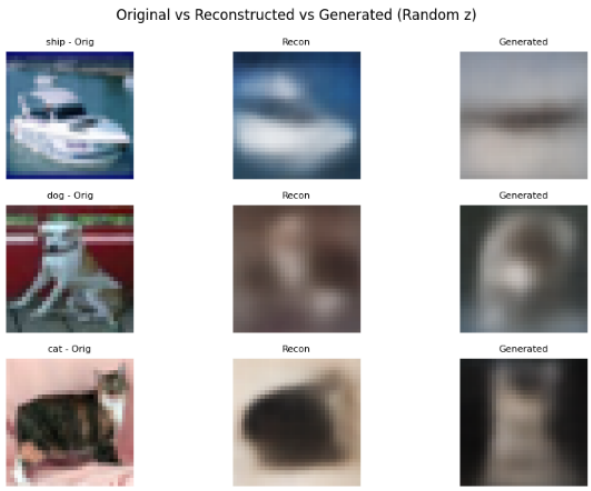

Figure 1: Original (left), reconstruction (middle), and generated image using $z \sim \mathcal{N}(0, I)$ (right).

2.2 Image Generation and Augmentation

After training, only the decoder was used to generate images. For each class, latent vectors $z \sim \mathcal{N}(0, I)$ were sampled and concatenated with the corresponding one-hot label before being passed through the decoder; encoder-derived vectors were excluded as they produce reconstructions rather than novel samples. Three augmentation levels (10, 50, and 100 generated images per class) were evaluated against a real-data baseline using the same CNN architecture, Adam optimiser (lr $= 5 \times 10^{-4}$), 30 training epochs, and a fixed stratified validation split of 5,000 samples.

3 Results

The results indicate that small-scale augmentation (+10/class) slightly degraded performance, likely due to the limited contribution of synthetic data relative to the original dataset. Moderate augmentation (+50/class) produced a marginal improvement, while the highest level (+100/class) achieved the best performance, improving accuracy by 0.81% over the baseline. Overall, the impact remained modest, suggesting that imperfect sample quality and limited quantity constrained the overall gain. Minor variation in baseline accuracy is attributable to random initialisation, data shuffling, and stochastic optimisation.

As shown in Figure 2, validation accuracy curves follow a consistent trajectory across all configurations, converging to approximately 70% by epoch 5 and remaining stable thereafter. The close alignment of all four curves confirms that synthetic augmentation did not meaningfully alter the learning dynamics, further supporting the conclusion that the scale and quality of generated samples were insufficient to shift decision boundaries.

Table 2: CNN Test Accuracy Under Each Augmentation Configuration

Configuration	**Accuracy**	**Δ Baseline**
Baseline	67.98%	—
Aug +10/class	67.28%	−0.70%
Aug +50/class	68.20%	+0.22%
Aug +100/class	68.79%	+0.81%

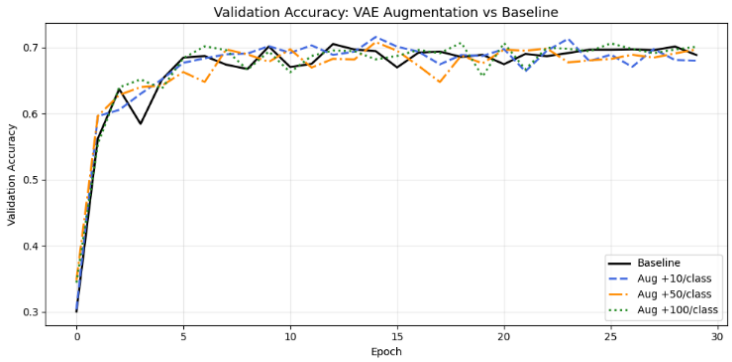

Figure 2: Validation accuracy across 30 epochs for the baseline and each augmentation configuration.

4 Observations

4.1 Analysis of Augmentation Performance

The experimental results show that CVAE-based augmentation leads to only modest improvements in classification performance. While higher augmentation levels (+100 samples per class) resulted in a small gain in accuracy, the overall impact remains limited. This behavior can be explained by several key factors.

1. **Scale mismatch.** Even at +100 samples per class (1,000 total), synthetic data represents a small fraction of the 45,000-image training set, limiting its influence on learned decision boundaries.
2. **Image quality.** Generated samples exhibit slight distortions and lack fine-grained detail. Since CNNs rely on high-frequency features such as edges and textures, these imperfections reduce the effectiveness of synthetic data as a training signal.
3. **Limited latent novelty.** Regularisation toward a standard normal prior enables diversity but constrains generation to the existing data manifold, introducing few new discriminative features.
4. **Augmentation scale.** Marginal gains at +50 and +100 per class suggest synthetic data becomes more meaningful at higher volumes, but sample quality limits the overall benefit.

4.2 When VAE Augmentation Could Help

Although no improvement was observed in this experiment, VAE-based augmentation can still be beneficial in low-data scenarios. When real samples are limited, synthetic data can enhance diversity and improve generalization. Thus, the negative results here are specific to the data-rich CIFAR-10 setting, not a general limitation of VAEs.

4.3 Motivation for Part (c)

The main limitation identified in this study is the poor visual quality of VAE-generated images. Generative Adversarial Networks (GANs) address this issue by using an adversarial objective, where a discriminator guides the generator to produce sharper and more realistic images. This suggests that GAN-based augmentation may provide more useful training data and lead to improved classification performance, which is explored in Part (c).